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Chapter 1

Using the Revision Mathematics Notes

Start Revision Maths by doing the quizzes which cover

- Quiz 1: Introduction to Vectors and Matrices.
- Quiz 2: Vectors
- Quiz 3: Matrices
- Quiz 4: Multivariate calculus
- Quiz 5: Introduction to Topology

The quizzes are diagnostic, designed to tell you where you have gaps. If you have no gaps in the quiz questions which are labelled either essential or useful you can move on directly to preparation for the other parts of EC400.

Quiz 1 covers basic material; if you are completely confident that you know this already skip the quiz. You should do quizzes 2, 3 and 4 online. You will then be able to download a document with answers. Skip quiz 5 unless you are an MRes student or wish to take EC487 (Advanced Microeconomics). EC487 is compulsory for MSc Econometrics and Mathematical Economics students; for other students EC487 is available only with permission of the instructor.

If you have difficulty with the quizzes read the relevant part of these notes. These are notes only: many results are stated without proof, although in the more important cases intuition is provided. If you had an unlimited amount of time it would be desirable to both have the intuition and understand the proof. Do not despise intuition. It is always helpful. It is easy to be overwhelmed by the amount and complexity of the maths you meet as a postgraduate student in economics. Trying to develop intuition is very helpful. Can you draw a diagram that shows what is happening? Can you explain in words what is going on?

Given the time pressures you are likely to be under, if you can handle the material in the quizzes it is a better use of your time to move on to preparation for the rest of EC400. In economists' language the marginal product of an hour

spent preparing for other parts of EC400 is greater than the marginal product of studying the proofs of the results given here.

If you need additional material you may find helpful:

Sydsaeter, H and P. Hammond, **Essential** Mathematics for Economic Analysis. Prentice Hall.

You can use any edition of Sydsaeter and Hammond. Be careful on the exact title, there is a related book Sydsaeter, K, P. Hammond, A. Seirstad and A. Strom, **Further** Mathematics for Economic Analysis which is not appropriate for revision maths.

Chapters 4, 6 and 7 of Sydsaeter and Hammond cover functions of a single variable, including calculus. Chapters 11 and 12 on multivariate calculus, and 15 and 16 on matrices cover the core material of revision maths.

Simon, C.P. and L. Blume, Mathematics for Economists, Norton

is the core text for maths for microeconomics. Chapters 2 - 5 cover background material on functions of a single variable, including calculus. Chapters 8 - 11 on matrices and 13 and 14 on multivariate calculus cover the core material of revision maths, but at greater depth than the treatment here.